



MATEMATICI 2

Curs 9

Triunghiularizarea matricilor

Fie $A \in \mathcal{M}_n(\mathbb{R})$ având polinom. caracteristic cu descomp. liniară peste $\mathbb{R}$, adică $\sigma(A) = \{\lambda_1, \dots, \lambda_p\}$, cu multiplicitățile algebrice respective $n_1, \dots, n_p$ și $n_1 + n_2 + \dots + n_p = n$.

În aceste cond., A e triunghiularizabilă, adică $\exists\, T \in \mathcal{M}_n(\mathbb{R})$ nesingulară aî. $T^{-1}AT$

$$T^{-1}AT = \begin{pmatrix} \lambda_1 & & & * \\ & \ddots & & \\ & & \lambda_1 & \\ & & & \ddots \\ & & & & \lambda_p \\ 0 & & & & & \ddots \\ & & & & & & \lambda_p \end{pmatrix}$$

(n_1 ori λ_1, ..., n_p ori λ_p; * oricе)

= formă superior triunghiulară

Dem.: Vom face dem. după n.

Cons. $P(n-1)$ adev. și dem $P(n)$.

Obs. $P(1)$ adev.

$\subseteq$ Fie $\lambda_1 \in \sigma(A) \Rightarrow \exists v_1 \in V_{\lambda_1}$, $v_1 \neq 0$.

Completăm v_1 până la o bază a lui $\mathbb{R}^n$ și

construim matricea $T_1 = (v_1 \; \underbrace{v_2 \; \dots \; v_n}_{n-1 \text{ vect.}})$

$$T_1^{-1} A T_1 = \begin{pmatrix} \lambda_1 & * & \dots & * \\ 0 & & & \\ \vdots & & A^* & \\ 0 & & & \end{pmatrix} = A_1$$

$$P_A(\lambda) = \det(A - \lambda I_n) = \det(A_1 - \lambda I_n) =$$

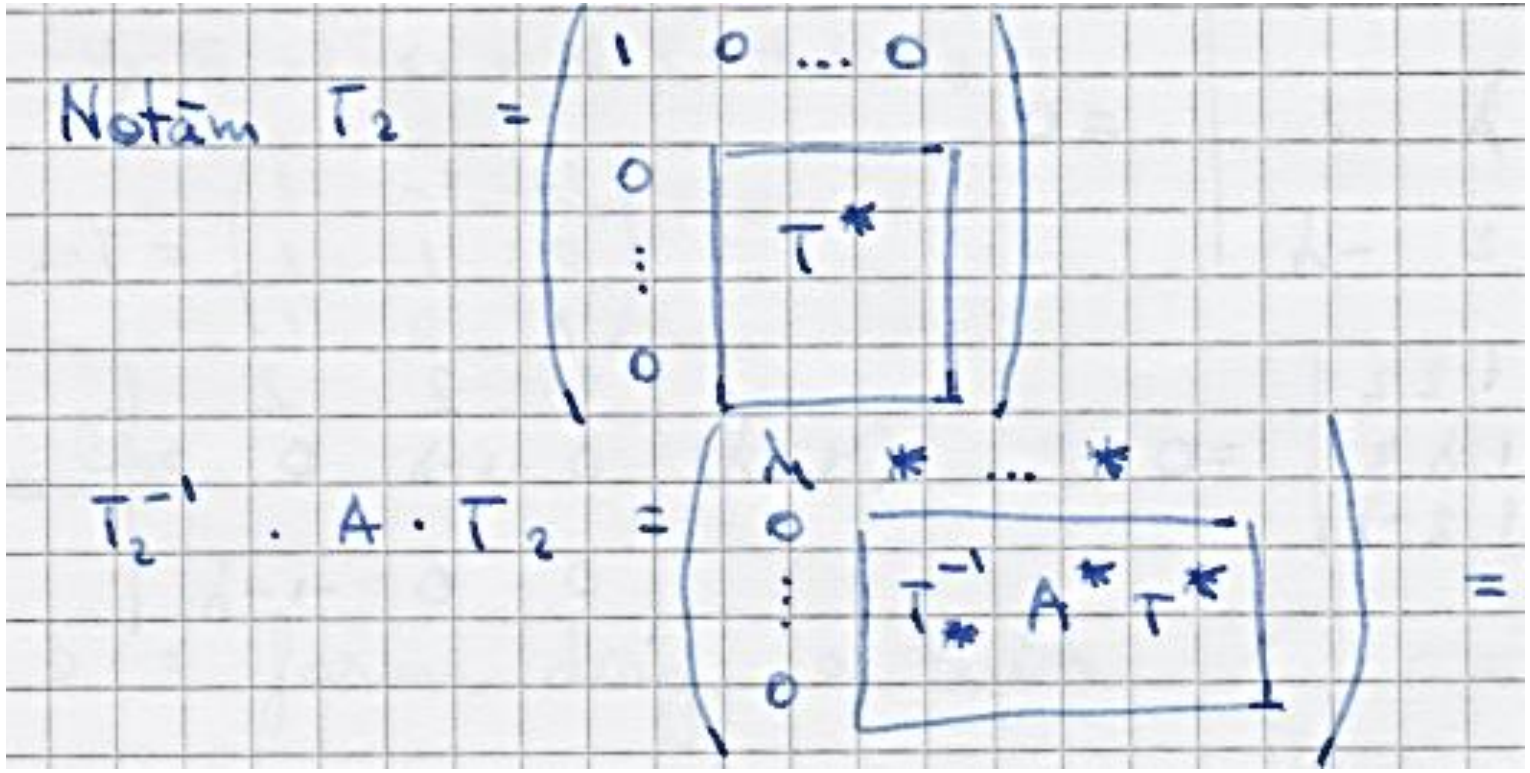
$$= (\lambda - \lambda_1) \det(A^* - \lambda I_{n-1}) = (\lambda - \lambda_1) P_{A^*}(\lambda)$$

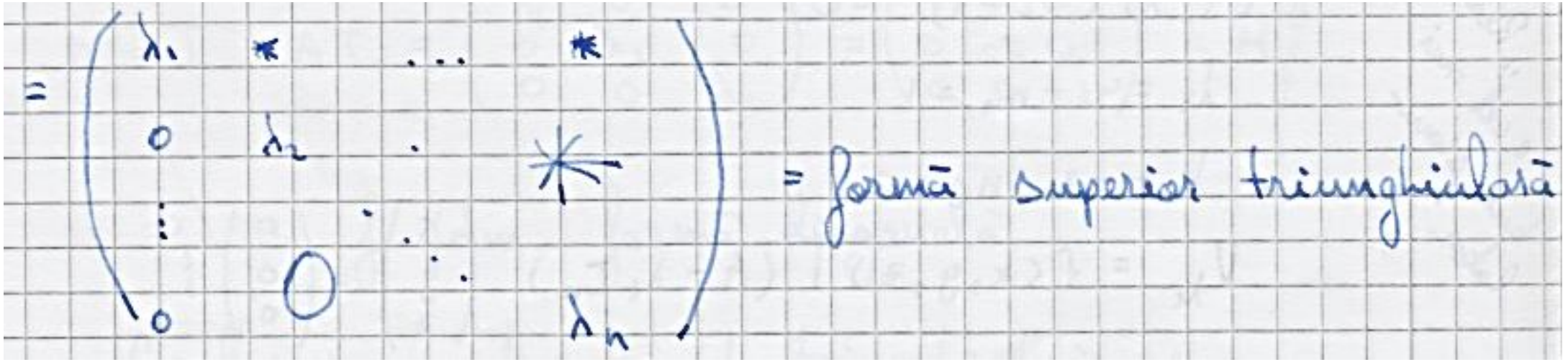
$P_A(\lambda)$ adm. descompunere liniară $\Rightarrow$

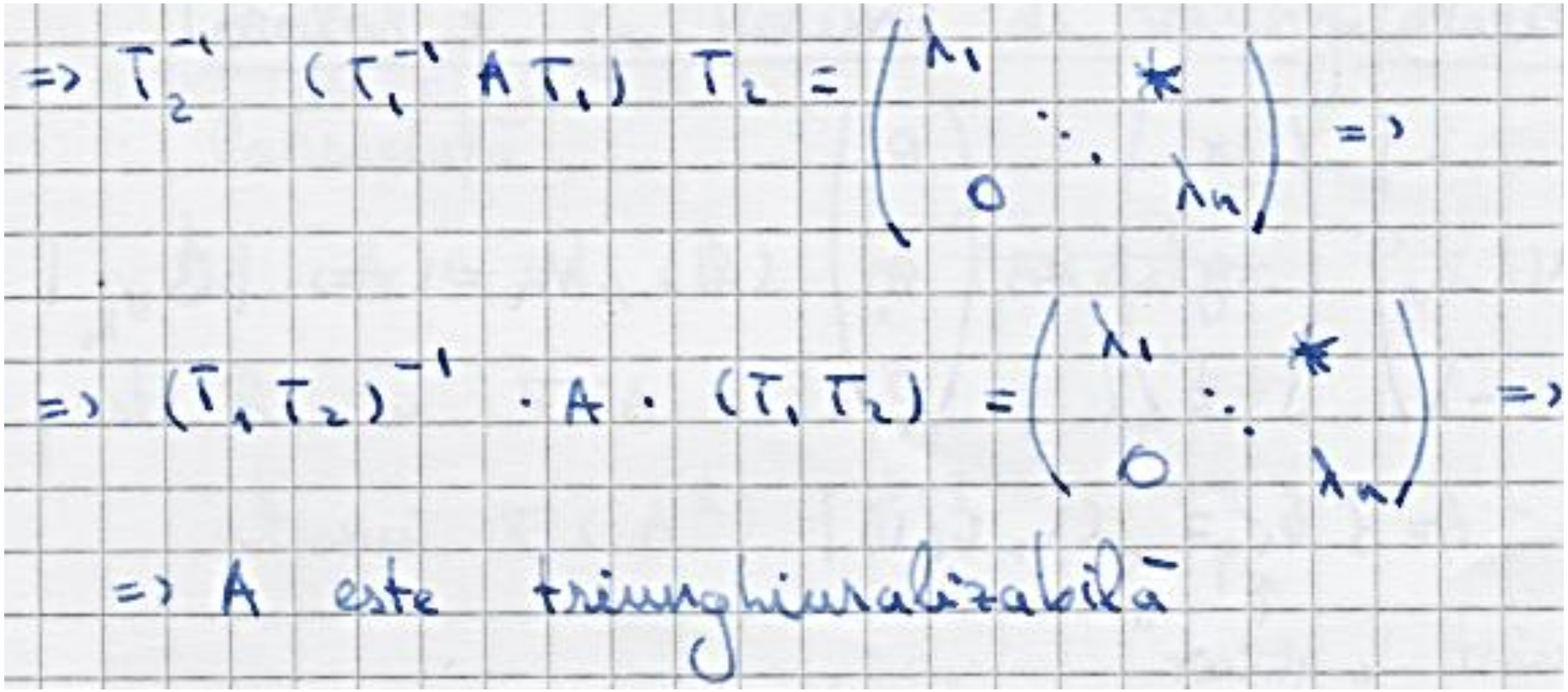
$\Rightarrow$ $P_{A_1}^*(\lambda)$ adm. descompunere liniară peste $\mathbb{R}$

$\Rightarrow$ $\exists\, P_A^*(\lambda)$ descomp. liniară $\Rightarrow$

$\Rightarrow$ $\exists\, T^* \in \mathcal{M}_{n-1}(\mathbb{R})$ a.î. $T^{*-1} A^* T^* = \begin{pmatrix} \lambda_2 & & * \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}$


Notăm $T_2 =$
$\begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & & & \\ \vdots & & T^* & \\ 0 & & & \end{pmatrix}$
$T_2^{-1} \cdot A \cdot T_2 =$
$\begin{pmatrix} \lambda & * & \dots & * \\ 0 & & & \\ \vdots & & T_*^{-1} A^* T^* & \\ 0 & & & \end{pmatrix} =$


= $\begin{pmatrix} \lambda_1 & * & \cdots & * \\ 0 & \lambda_2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & * \\ 0 & \cdots & 0 & \lambda_n \end{pmatrix}$ = formă superior triunghiulară


$\Rightarrow T_2^{-1} (T_1^{-1} A T_1) T_2 = \begin{pmatrix} \lambda_1 & & * \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix} \Rightarrow$
$\Rightarrow (T_1 T_2)^{-1} \cdot A \cdot (T_1 T_2) = \begin{pmatrix} \lambda_1 & & * \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix} \Rightarrow$
$\Rightarrow$ A este triunghiularizabilă

ex.: Să se diagonalizeze $f : \mathbb{R}^3 \to \mathbb{R}^3$,

$f(x_1, x_2, x_3) = (2x_2 + 2x_3,\ 2x_1 + 2x_3,\ 2x_1 + 2x_2)$

Rez.: $B_c = \{e_1 = (1,0,0),\ e_2 = (0,1,0),\ e_3 = (0,0,1)\}$

$f(e_1) = f(1,0,0) = (0,2,2) = 0 \cdot e_1 + 2 \cdot e_2 + 2 \cdot e_3$

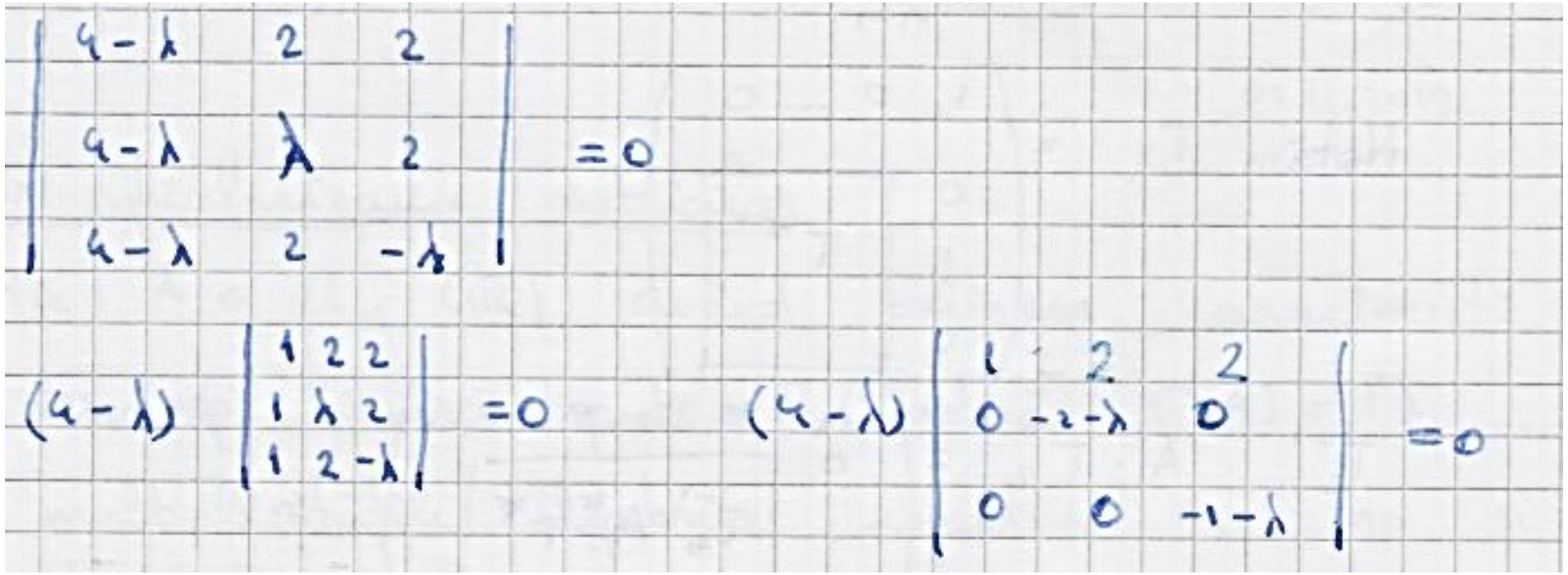
$f(e_2) = f(0,1,0) = (2,0,2) = 2 \cdot e_1 + 0 \cdot e_2 + 2 \cdot e_3$

$f(e_3) = f(0,0,1) = (2,2,0) = 2 \cdot e_1 + 2 \cdot e_2 + 0 \cdot e_3$

$$M_f^{\mathcal{B}_c} = \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 0 \end{pmatrix}$$

$$P_f(\lambda) = 0 \Leftrightarrow \det(A - \lambda I_3) = 0 \Leftrightarrow$$

$$\Leftrightarrow \begin{vmatrix} 0-\lambda & 2 & 2 \\ 2 & 0-\lambda & 2 \\ 2 & 2 & 0-\lambda \end{vmatrix} = 0$$


$\begin{vmatrix} 4-\lambda & 2 & 2 \\ 4-\lambda & \lambda & 2 \\ 4-\lambda & 2 & -\lambda \end{vmatrix} = 0$
$(4-\lambda)\begin{vmatrix} 1 & 2 & 2 \\ 1 & \lambda & 2 \\ 1 & 2 & -\lambda \end{vmatrix} = 0$
$(4-\lambda)\begin{vmatrix} 1 & 2 & 2 \\ 0 & -2-\lambda & 0 \\ 0 & 0 & -1-\lambda \end{vmatrix} = 0$

$$(4-\lambda)(-2-\lambda)^2 = 0$$

$\lambda_1 = 4 \quad n_1 = 1$

$\lambda_2 = -2 \quad n_2 = 2$

$$V_{\lambda_i} = \{(x,y,z) \mid (A - \lambda_i I_n) \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}\}$$

$$\begin{pmatrix} -\lambda & 2 & 2 \\ 2 & -\lambda & 2 \\ 2 & 2 & -\lambda \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$n_1 = 1 \Rightarrow |B_{V_{\lambda_1}}| = 1$

$B_{V_{\lambda_1}} = \{ v_1 = (1, 1, 1) \}$

↑

„ghicit"

$B\ V_{\lambda_2}$

$n_2 = 2$

$$\begin{pmatrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$2x + 2y + 2z = 0$

$y = \alpha, \quad z = \beta$

$x = -\alpha - \beta$

$(x, y, z) = (-\alpha - \beta, \alpha, \beta) = \alpha \underbrace{(-1, 1, 0)}_{v_2} + \beta \underbrace{(-1, 0, 1)}_{v_3}$

$B_{V_{1,2}} = \{ v_2 = (-1,1,0),\ v_3) = (-1,0,1) \}$

$B = \{ v_1 = (1,1,1),\ v_2 = (-1,1,0),\ v_3 = (-1,0,1) \}$

$v_1 = 1 \cdot e_1 + 1 \cdot e_2 + 1 \cdot e_3$

$v_2 = -1 \cdot e_1 + 1 \cdot e_2 + 0 \cdot e_3$

$v_3 = -1 \cdot e_1 + 0 \cdot e_2 + 1 \cdot e_3$

$$T = \begin{pmatrix} 1 & -1 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

Obs.:

D = forma diag. a lui A

$$D = T^{-1} A T = \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_2 \end{pmatrix} = \begin{pmatrix} 4 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{pmatrix} = M_f^{\mathcal{B}}$$

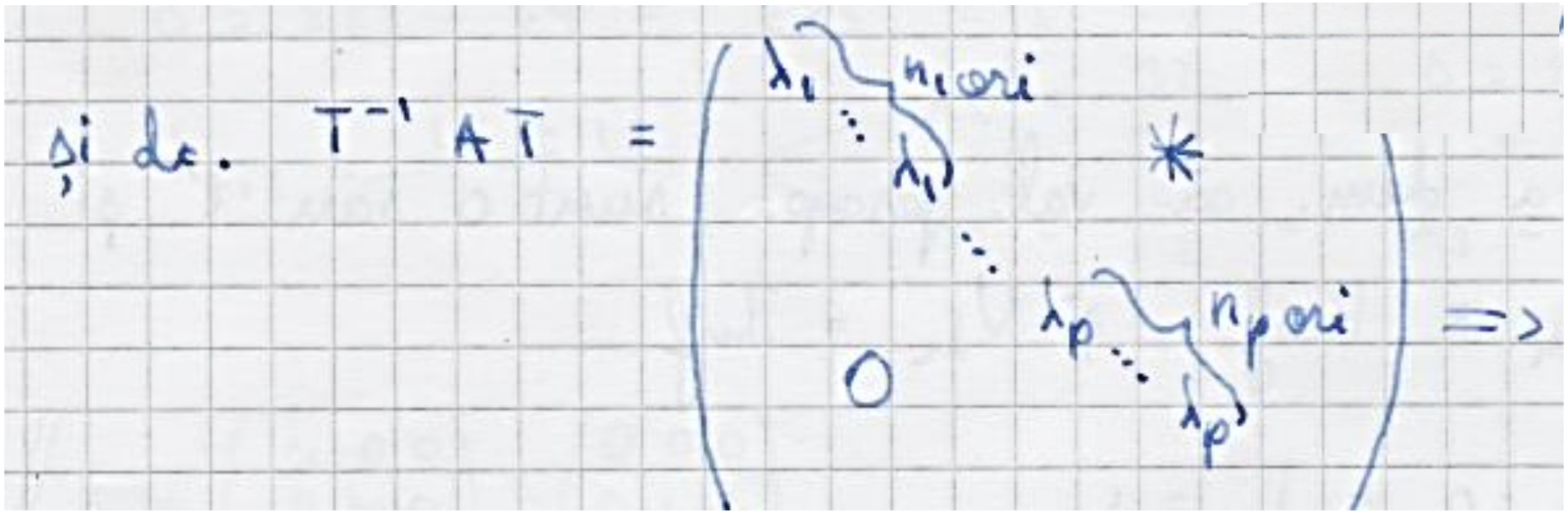
$\Rightarrow$ f adm. formă diagonală

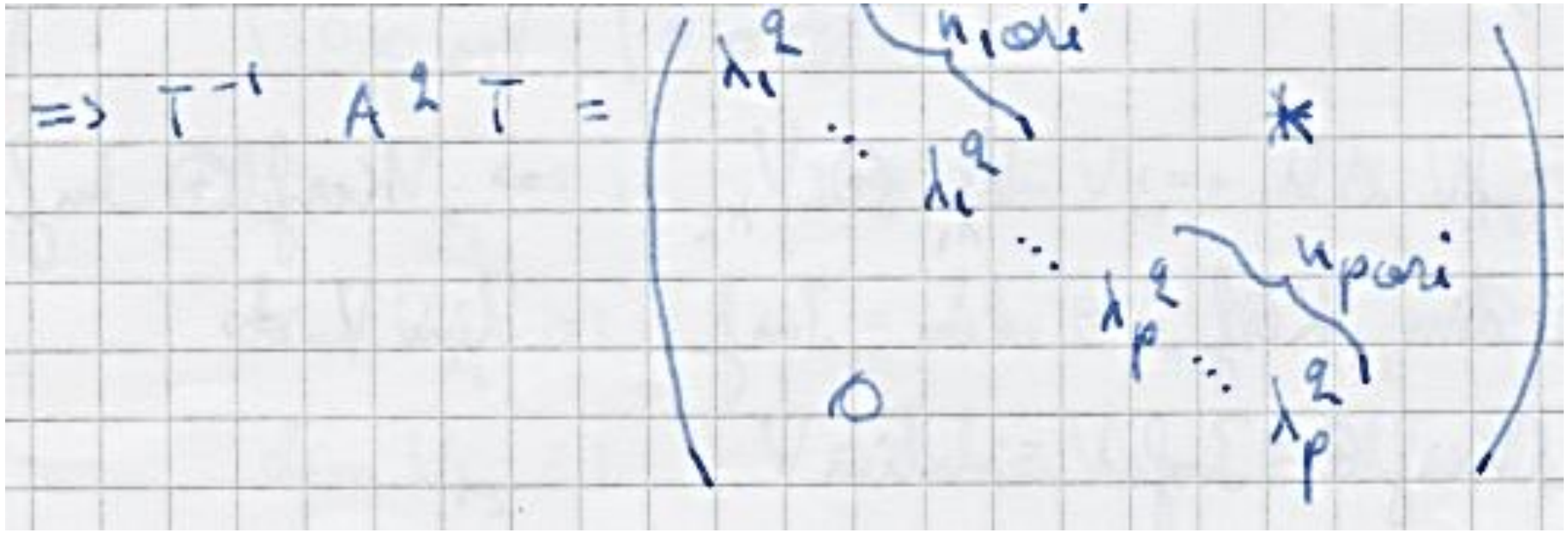
Corolar la teorema de triunghiularizare

Consecința 1 :

Dc. $A \in \mathcal{M}_n(\mathbb{R})$ și $P_A(\lambda)$ = descomp. liniară peste $\mathbb{R}$ cu $\sigma(A) = \{\lambda_1, \ldots, \lambda_p\}$,

atunci $\sigma(A^2) = \{\lambda_1^2, \ldots, \lambda_p^2\}$, A^2 adm. formă sup. (și) triunghiulară


$T^{-1}AT =$
λ_1
n_1 ori
λ_1
*
λ_p
n_p ori
λ_p
0
=>

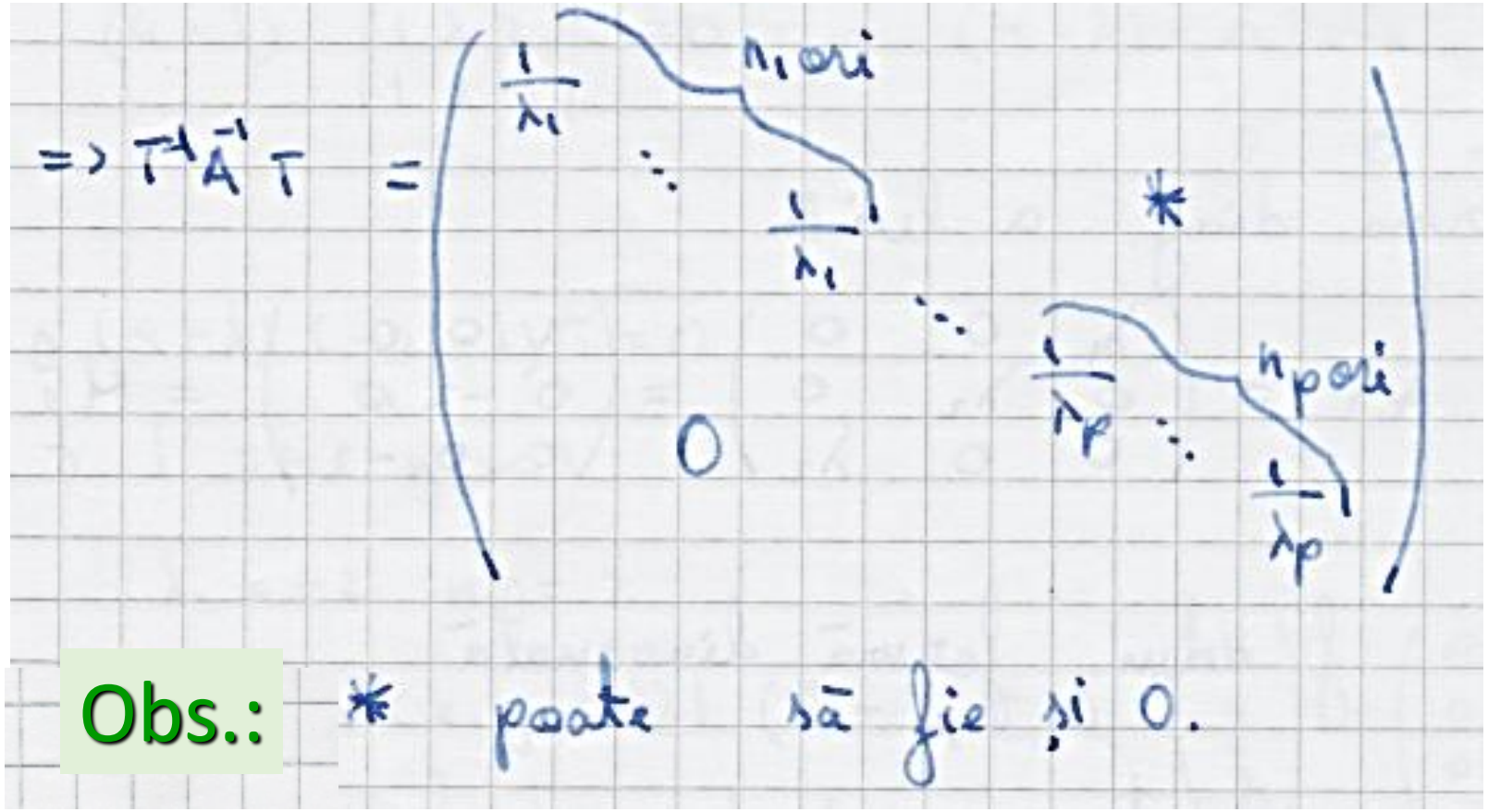

$\Rightarrow T^{-1} A^2 T =$
λ_1^2
n_1 ori
λ_1^2
$*$
λ_p^2
n_p ori
λ_p^2
λ_p^2
0

Consecința 2 :

Dc. $A \in \mathcal{M}_n(\mathbb{R})$, val. prop. sunt nenule ($\neq 0$) și polinom. caract. cu descomp. liniară peste $\mathbb{R}$,

atunci A^{-1} adm. formă sup. Δ

și dc. $T^{-1}AT = \begin{pmatrix} \lambda_1 & & & * \\ & \ddots & & \\ & & \lambda_p & \\ 0 & & & \ddots \\ & & & & \lambda_p \end{pmatrix}$ ⇒



Obs.: * poate să fie și 0.